

03_UI_UX_Spec.md

MonitorAI v1.5 UI / UX Specification

Version: 1.5

Status: Baseline

1. Design Philosophy

The user interface exists to help patients create standardized clinical records.

The interface should:

- Reduce confusion
- Reduce cognitive load
- Improve completion rate
- Improve image quality

The interface should never prioritize visual effects over usability.

2. Orientation

Supported Orientation:

Landscape Only

Portrait Mode:

Not Supported

All screens must be optimized for landscape operation.

3. Global Screen Layout

Display Order:

Camera Preview



Overlay Guide



PTG Layer



Status Layer



Voice Guidance



Haptic Feedback

4. Position Calibration Screen

Purpose:

Align the opening of the Invisalign Lens (Retractor Opening) to the center of the screen.

Display Elements

Live Camera Preview

3 × 3 Grid

Center Marker

Opening Detection Overlay

Instruction Label

Progress Indicator

Instruction

撮影口を画面中央に合わせてください

Success State

Opening Center Error < 5%

AND

Stable 3 Seconds

↓

Automatic Advance

Feedback

Voice:

撮影口を画面中央に合わせてください

Haptic:

Short Confirmation Tap

5. Capture Screen Layout

Used For:

Step01-Step11

Main Elements

Live Camera Preview

Overlay PNG

PTG Rail

Step Information

Progress Indicator

Zoom Selector

Status Message

6. Overlay PNG Layer

Opacity:

50%

Default

Adjustable

Range:

30–70%

Overlay Purpose

Provide positioning guidance.

The overlay is not used for scoring.

The overlay is not used for diagnosis.

7. PTG Rail

Position:

Right Side

Vertical

Display

Dot1

Dot2

Dot3

Dot4

Dot5

Dot6

Meaning

Dot1

Poor

Dot2

Improving

Dot3

Acceptable

Dot4

Minimum Passing

Dot5

Recommended

Dot6

Ideal

Animation

Active Dot:

Highlighted

Transition:

Smooth

No flashing effects.

8. Step Information

Display:

Current Step

Current Group

Short Instruction

Example:

Step02

Right Buccal ON

顔を右へ向けてください

9. Progress Indicator

Display:

Current Step / Total Steps

Example:

Progress Bar

Visible

Always Displayed

10. Zoom Selector

Location:

Bottom Right

Options

Normal

2.5x

Large

3.0x

Extra Large

4.0x

Important

Preview Only

Stored Image:

Always 1.0x

11. Status Messages

Examples:

位置を調整してください

↓

もう少し奥歯を見せてください

↓

良い位置です

↓

撮影します

Status messages should be short and clear.

12. Voice Guidance UX

Voice guidance is enabled by default.

Purpose:

Reduce the need to watch the screen continuously.

Events

Calibration

Dot4

Dot5

Dot6

Plateau

Capture Complete

Session Complete

13. Haptic UX

Purpose:

Provide progress feedback without visual attention.

Events

Calibration Complete

Dot4

Dot5

Dot6

Capture Complete

Upload Complete

14. Auto Capture Feedback

When capture occurs:

Visual:

Capture Flash Animation

Voice:

撮影しました

Haptic:

Strong Success Pulse

15. Review Screen

Displayed after Step11.

Display

11 Thumbnail Images

Capture Status

Retake Buttons

Upload Button

Retake

Individual Image Retake Only

Full Session Retake Not Required

16. Upload Screen

Display:

Upload Progress

Current File

Remaining Files

Connection Status

Progress Example

Uploading

7 / 11

17. Completion Screen

Displayed after successful upload.

Message

撮影が完了しました

ご協力ありがとうございました

Actions

Close

Return Home

18. Accessibility

Support:

- Dynamic Text
- Voice Guidance
- Haptic Guidance

The workflow should remain usable even when the user is not continuously looking at the screen.

19. Error States

Examples:

Calibration Failed

Capture Failed

Upload Failed

Network Unavailable

Error UX

Explain:

What happened

Why it happened

How to recover

Avoid technical language.

20. Final UI Principle

The UI should feel:

Simple

Predictable

Guided

The user should never wonder:

"What do I do next?"

The application should always provide the next action.